MIL-L-27639C(USAF)
8 March 1967
SUPERSEDING
MIL-L-27639B(USAF)
16 November 1966

MILITARY SPECIFICATION

LIFE PRESERVER INFLATING ASSEMBLY FLU-1/P

1. SCOPE

1.1 This specification covers one type of life preserver inflating assembly designated FLU-1/P.

2. APPLICABLE DOCUMENTS

*2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard
PPP-B-636	Box, Fiberboard
PPP-B-676	Boxes, Set-up

Military

MIL-P-116	Preservation, Methods of
MIL-C-25369	Cylinder, Carbon Dioxide, for Inflation of Pneumatic Life Preservers

STANDARDS

Federal

FED-STD-151	Metals; Test Methods

Military

MIL-STD-100	Engineering Drawings and Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for The Selection of
MS33586	Metals, Definition of Dissimilar

DRAWINGS

Air Force

52A6600	Cap - Inflator Valve, Pneumatic Life Preserver
58A3568	Gasket - Life Preserver, Inflator Stem, Upper
58A3569	Gasket - Life Preserver, Inflator Stem, Lower
61B4286	Stem - Inflator Valve, Life Preserver Cell Assembly of
61B4313	Lanyard, Valve - Life Preserver

(Copies of specifications, standards, drawings, and publications required by

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suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification includes provisions for preproduction testing (see 4.4 and 6.2).

* 3.2 Components. The life preserver inflating assembly shall consist of the following major components:

Item No.	Quantity	Item Name
1	1	Inflator (see 3.5.1)
2	2	Outlet gaskets (see 3.5.2)
3	1	Inflator valve cap (see 3.5.3).

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Age. Unless otherwise specifically approved by the procuring activity, all materials and component parts used in the life preserver inflating assembly shall have been manufactured not more than 12 months prior to the date of delivery to the Government.

3.4.2 Metals. Metals shall be the corrosion-resistant type or suitably treated to resist corrosion due to salt spray or atmospheric conditions likely to be met in storage or normal service.

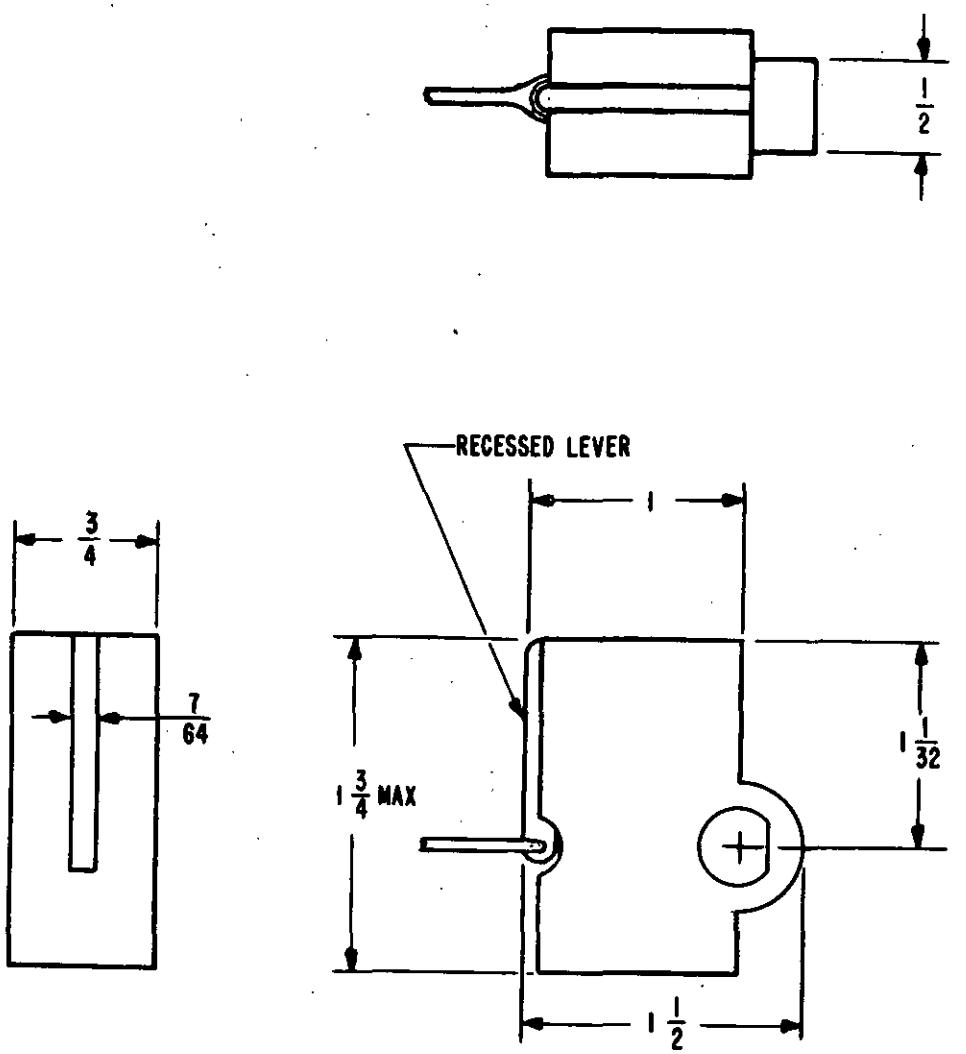
3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

* 3.5 Design

* 3.5.1 Inflator. The envelope or the finished external dimensions of the inflator shall conform to figure 1. The inflator shall consist of a housing, an internal piercing pin, and an external actuating lever.

* 3.5.1.1 Housing. The housing shall be constructed of high pressure die-cast aluminum which has been anodized and treated with a dichromate sealer. The housing shall have an outlet and an inlet. The inlet shall be threaded to accept a carbon dioxide cylinder conforming to type II of MIL-C-25369. A gasket shall be recessed in the inlet to provide a gas tight seal when a carbon dioxide cylinder is installed. A screw type lock shall be provided to retain the installed carbon dioxide cylinder. The screw head and slot shall be rounded and free from burrs and shall not protrude above the housing when tightened against the cylinder. The top of the tapped hole shall be countersunk, smooth, and free from burrs or sharp edges. The outlet shall be designed to fit on an inflator valve stem, conforming to Drawing 61B4286, and so that the gas passages will align with the hole in the side of the inflator valve stem regardless of which side of the outlet is toward the base of the inflator valve stem. The housing shall have a vent passage for venting any residual air, which may be trapped in the life preserver cell, from the outlet through the housing before the carbon dioxide cylinder is punctured. The housing shall also have passages for conducting the gas from the punctured carbon dioxide cylinder to the outlet.

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UNLESS OTHERWISE SPECIFIED,
DIMENSIONS ARE IN INCHES.
TOLERANCE ON FRACTIONS ± 1/64

*FIGURE 1. Inflator

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* 3.5.1.1.1 Piercing pin. The piercing pin shall be constructed of cadmium-plated hardened steel, having a case depth of 0.010 inch. It shall have a spring to retract the pin immediately from the opening of the punctured carbon dioxide cylinder. A seal shall be provided to prevent leakage of the pressure of the life preserver cell after the piercing pin has been retracted.

3.5.1.1.2 Actuating lever. The lever shall be constructed of cadmium-plated hardened steel that is 7/64 inch thick and has a case depth of 0.010 inch. All edges of the lever shall have a minimum radius of 0.010 inch. The lever shall be held in position with a staked corrosion-resistant-steel pin. In the unoperated position the lever shall be recessed into the housing. A stop shall be provided to prevent the lever from traveling a distance of more than 90 degrees from the horizontal axis of the housing. A means shall be provided to return the lever to within 10 degrees from the unoperated position immediately after discharge of the carbon dioxide cylinder. A portion of the lever shall be colored red to indicate that the vent has been closed and locked. A lanyard conforming to Drawing 61B4313 shall be attached to the free end of the lever.

* 3.5.2 Outlet gaskets. Gaskets conforming to Drawings 58A3568 and 58A3569 shall be provided to effect a seal at the outlet.

* 3.5.3 Inflator valve cap. An inflator valve cap conforming to Drawing 52A6600 shall be provided to hold the housing assembly in place on the inflator valve stem.

3.6 Performance

* 3.6.1 Operating force without cylinder. The life preserver inflating assembly shall operate with a load of not more than 2 pounds attached to the lanyard end of the actuating lever.

* 3.6.2 Cylinder installation and lock. A cylinder conforming to type II of MIL-C-25369 installed with a torque of 20 inch-pounds shall compress the inlet gasket and bottom out providing a leaktight seal. With the cylinder lock engaged, a torque of not less than 30 inch-pounds nor more than 60 inch-pounds shall be required to remove the cylinder. The piercing pin shall not contact the installed cylinder while the lever is in the unoperated position.

* 3.6.3 Vent flow. A vent flow of at least 3.0 liters per minute shall be provided when the lever is in the unoperated position.

* 3.6.4 Vent lock and leakage. The vent shall close and lock before the piercing pin contacts the cylinder. The lever shall return to a position not more than 10 degrees from the unoperated position, and the vent shall remain closed. It shall not be possible to return the lever to the unoperated position without resetting the vent to the open position. The vent shall not leak when closed and a pressure of 40 pounds per square inch (psi) is applied to the outlet.

* 3.6.5 Operation. An installed cylinder conforming to type II of MIL-C-25369 shall be punctured and discharged within 2-1/2 seconds when a force of 20 pounds causes the lever to travel a maximum arc of 90 degrees from the longitudinal axis of the life preserver inflating assembly. The lever shall automatically return to a position not more than 10 degrees from the unoperated position when released.

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* 3.6.6 Hydrostatic pressure. The life preserver inflating assembly shall satisfactorily withstand a hydrostatic pressure of 1500 psi without deformation or leakage.

* 3.6.7 Torque. The life preserver inflating assembly shall withstand a minimum torque of 100 inch-pounds while being attached to a carbon-dioxide cylinder conforming to type II of MIL-C-25369.

* 3.6.8 Salt spray. The life preserver inflating assembly shall pierce and discharge, in not more than 2-1/2 seconds, a cylinder conforming to type II of MIL-C-25369 with a load of 20 pounds on the lever arm after subjection to salt spray for 96 hours as specified herein.

* 3.6.9 Reliability. The life preserver inflating assembly shall have a minimum inherent reliability of 0.9990 at a confidence level of 95 percent.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

* 3.8 Identification of product. The life preserver inflating assembly shall be marked for identification in accordance with MIL-STD-130 and shall include the following specific information:

Life Preserver Inflating Assembly
FLU-1/P
MIL-L-27639C(USAF)
Manufacturer's Identification*
Contract or Order No.*
Date of Manufacture*

*The manufacturer shall insert the applicable information.

* 3.9 Workmanship. The life preserver inflating assembly, including all parts and equipment, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship shall be of the highest quality to insure proper functioning of the life preserver inflating assembly at all times.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of life preserver inflating assemblies shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

4.3 Test conditions

4.3.1 Atmospheric conditions. Unless otherwise specified herein, all tests required by this specification shall be made at an atmospheric pressure of 28 to 38 inches of mercury, at a temperature of 70 (plus or minus 18) degrees Fahrenheit, and at a relative humidity of 80 percent or less. Where tests are concluded with atmospheric pressure or temperature different from the starting values, corrections shall be made for the change in values.

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*4.3.2 Installed cylinders. The cylinders shall be installed with a torque of 20 (plus 10 or minus 0) inch-pounds. Discharged cylinders shall be installed in life preserver inflating assemblies tested for hydrostatic pressure.

4.4 Preproduction testing

*4.4.1 Preproduction test samples. The preproduction test samples shall consist of 36 life preserver inflating assemblies that have been tested as specified in 4.4.2. The preproduction test samples shall be representative of the production equipment and shall be identified with the manufacturer's part number and any other information required by the procuring activity (see 6.2).

*4.4.2 Preproduction tests. Six of the preproduction test samples specified in 4.4.1 shall be subjected to the tests described in 4.6.1 through 4.6.9. Thirty of the untested preproduction test samples specified in 4.4.1 shall be subjected only to the reliability demonstration specified in 4.6.10, without failure.

*4.4.3 Preproduction test data. Preproduction test data, including reliability demonstration, shall be in accordance with the instructions of the procuring activity (see 6.2).

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests.

*4.5.1 Individual tests. Each life preserver inflating assembly shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Operations test without cylinder

c. Cylinder installation and lock

d. Vent flow

e. Vent lock and leakage

f. Operation.

*4.5.2 Sampling tests

*4.5.2.1 Lot. A lot shall consist of life preserver inflating assemblies manufactured under essentially the same conditions and submitted for inspection at substantially the same time. A lot shall not exceed 500 life preserver inflating assemblies for sampling plan A and shall be in accordance with inspection level S-3 of MIL-STD-105 for sampling plan B.

*4.5.2.2 Plan A. One life preserver inflating assembly selected at random from the first lot, and from each fourth lot thereafter, shall be subjected to the following tests as described under 4.6:

a. Hydrostatic pressure

b. Torque

c. Salt spray.

*4.5.2.3 Plan B. Upon completion of the individual tests specified in 4.5.1, random samples of life preserver inflating assemblies shall be selected from each lot in accordance with inspection level S-3 of MIL-STD-105. The sample size shall be based only on the applicable size code letter corresponding to the inspection level S-3. Each life preserver inflating assembly selected as a sample unit shall be forwarded to the inspection activity selected by the procuring activity (see 6.2 and 6.3). The specified inspection activity shall subject the life preserver inflating assemblies to the following tests as described under 4.6:

a. Examination of product

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b. Operations test without cylinder

c. Cylinder installation and lock

d. Vent flow

e. Vent lock and leakage

f. Operation.

*4.5.2.4 Rejection and retest (plan A only). When one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure have been determined. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After corrections, all sampling tests shall be repeated.

*4.5.2.5 Rejection (plan B only). Rejection and acceptance for plan B shall be as follows: The acceptance number shall be zero, and the rejection number shall be one for each test. The lot from which the life preserver inflating assembly samples were selected shall not be shipped until the samples have been approved by the procuring activity and the lot has been accepted by the cognizant quality control representative. The lot from which the samples were selected shall be rejected if any sample fails to meet any requirement of the tests specified in plan B. A rejected lot shall not be resubmitted except with the approval of the procuring activity.

*4.5.2.6 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. Final acceptance of the entire lot or lots produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the methods of correcting them.

4.6 Test methods

*4.6.1 Examination of product. Each life preserver inflating assembly shall be visually examined to determine compliance with the requirements of this specification with respect to workmanship, materials, marking, and construction. Each life preserver inflating assembly shall be checked to determine if the inlet is threaded to the required depth.

*4.6.2 Operations test without cylinder. Each life preserver inflating assembly shall be held in an inverted position with a 2-pound load attached to the end of the lever arm and the lever arm held in the normal position. When the lever arm is released, the load shall operate the piercing mechanism.

*4.6.3 Cylinder installation and lock. The actuating lever shall be in the unoperated position. A cylinder conforming to type II of MIL-C-25369 shall be installed with a torque of 20 inch-pounds. The cylinder shall compress the inlet gasket and bottom out against the metal in the inlet. The cylinder lock shall be engaged and the cylinder shall be loosened by applying a counterclockwise torque of not less than 30 inch-pounds nor more than 60 inch-pounds. The lock shall then be disengaged and the cylinder shall be removed and examined. Any indication of contact with the piercing pin shall be cause for rejection.

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* 4.6.4 Vent flow. The lever shall be in the unoperated position. The life preserver inflating assembly shall be mounted on an inflator valve stem conforming to Drawing 61B4286 with the gaskets and the inflator valve cap in place. A cylinder conforming to type II of MIL-C-25369 shall be installed in the inlet and a pressure of 2 psi shall be applied to the outlet. The vent flow shall be recorded. A flow of less than 3.0 liters per minute shall be cause for rejection.

*4.6.5 Vent lock and leakage. A cylinder conforming to type II of MIL-C-25369 shall be installed in the inlet of the life preserver inflating assembly. The actuating lever shall be operated through an arc sufficient to close and lock the vent passage. The cylinder shall not discharge. The lever shall be released and shall return to a position not more than 10 degrees from the unoperated position. The colored indicator on the lever shall be visible. A pressure of 40 psi shall be applied to the outlet. Failure of the vent to remain locked or any indication of leakage shall be cause for rejection. A method to determine vent closure and leakage during this test shall be devised by the manufacturer.

*4.6.6 Operation. A cylinder conforming to type II of MIL-C-25369 shall be installed in the life preserver inflating assembly. A force of 20 pounds shall be applied to the lever which shall cause it to travel in an arc from the unoperated position to its maximum position and to be released. Failure of the piercing pin to puncture the cylinder, retract, and discharge the gas in 2-1/2 seconds or travel of the lever in excess of 90 degrees from the longitudinal axis of the inflating assembly shall be cause for rejection. The lever shall automatically return to a position not more than 10 degrees from the unoperated position upon release.

*4.6.7 Hydrostatic pressure. With the lever in the vent lock position, the life preserver inflating assembly shall be subjected to a hydrostatic pressure of 1500 psi to the outlet for a period of not less than 5 minutes. At the end of this time a pressure of 40 psi shall be applied at the outlet for not less than 10 seconds. Any indication of leakage shall be cause for rejection.

*4.6.8 Torque. A cylinder conforming to type II of MIL-C-25369 shall be installed with a torque of 100 inch-pounds after which 40 psi shall be applied to the outlet. Any indication of leakage or fracture of the life preserver inflating assembly shall by cause for rejection.

*4.6.9 Salt spray. The life preserver inflating assembly shall be subjected to a salt spray test for 96 hours in accordance with method 811 of FED-STD-151. After completion of the salt spray test, a cylinder conforming to type II of MIL-C-25369 shall be installed in the life preserver inflating assembly. The life preserver inflating assembly shall then be held in an inverted position with a 20-pound load attached to the end of the lever arm and the lever arm held in the normal position. When the lever arm is released, the load shall operate the piercing mechanism which shall pierce and discharge the cylinder in not more than 2-1/2 seconds. There shall be no corrosion of the base metals of life preserver inflating assembly after the salt spray test.

* 4.6.10 Reliability demonstration. The life preserver inflating assembly shall be subjected to 100 cycles of reliability testing. A cycle shall consist of subjection

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of the life preserver inflation assembly to the tests specified in 4.6.4, 4.6.5, and 4.6.6. Failure of the inflating assembly to accomplish all of the cycles successfully shall be cause for rejection.

5. PREPARATION FOR DELIVERY

*5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

*5.1.1 Level A. Unless otherwise specified in the contract or order, each life preserver inflating assembly shall be individually preserved and packaged in accordance with method III of MIL-P-116 in a unit container conforming to PPP-B-566 or PPP-B-676.

*5.1.2 Level C. Unless otherwise specified in the contract or order, each life preserver inflating assembly shall be preserved and packaged to prevent deterioration and damage from the supply source to the initial receiving activity.

*5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

*5.2.1 Level A. Life preserver inflating assemblies preserved and packaged as specified in 5.1.1 shall be packed in exterior overseas type shipping containers conforming to PPP-B-636. Insofar as practical, exterior shipping containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and contain identical quantities. Containers shall be closed and strapped in accordance with the box specification and the appendix thereto.

*5.2.2 Level B. Level B packing shall be the same as level A packing except that domestic type exterior containers shall be used.

*5.2.3 Level C. Packages shall be placed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification Rules or regulations of any other common carriers as applicable to the mode of transportation.

*5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

*6.1 Intended use. The life preserver inflating assembly covered by this specification is intended to be used as a component of the LPU-3/P and the LPU-10/P life preservers.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Preproduction requirements (see 3.1 and 4.4.1)

c. Requests for results of preproduction tests including reliability demonstration (see 4.4.3)

d. Selection of inspection activity for sampling plan B (see 4.5.2.3)

e. Selection of applicable levels of preservation and packaging and packing (see 5.1 and 5.2).

*6.3 Test samples. The test samples specified in 4.5.2.3 should be returned to the manufacturer.

6.4 Annotation of changes. The margins of this specification are marked with an

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asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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